AI BASED REMOTE HEALTH MONITORING AND ALERT GENERATION FOR ELDERLY AND HIGH RISK PATIENTS

ABSRACT

This invention presents an AI-enabled remote health monitoring and alert generation system designed for elderly and high-risk patients. The system utilizes a wearable multi-sensor platform to continuously monitor physiological parameters such as heart rate, SpO_2, and temperature, along with motion data using an accelerometer–gyroscope module for fall and anomaly detection. An ESP32 microcontroller enables real-time data acquisition and secure transmission to a cloud-based infrastructure.

The system incorporates an intelligent framework that performs continuous analysis of physiological data using an anomaly detection model to identify abnormal patterns. In addition, a dedicated heart-related disease prediction module analyzes combined parameters to detect conditions such as bradycardia with hypothermia and hypertensive crisis with tachycardia. This prediction is achieved using an XGBoost model trained on a synthetically generated dataset created through a Gaussian Copula-based approach, preserving interdependencies among physiological signals.

The framework adapts to individual health trends using real-time and historical data, improving accuracy and reducing false alarms compared to conventional threshold-based systems. The trained model is containerized and deployed on a cloud-based inference server, with Firebase enabling real-time synchronization between wearable devices and cloud services.

The system supports automated alerts, real-time communication with caregivers, and mobile application-based monitoring with multi-user support through UID-based mapping. By integrating wearable sensing, machine learning, and cloud intelligence, the invention enables early detection of critical conditions and supports proactive healthcare management.

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